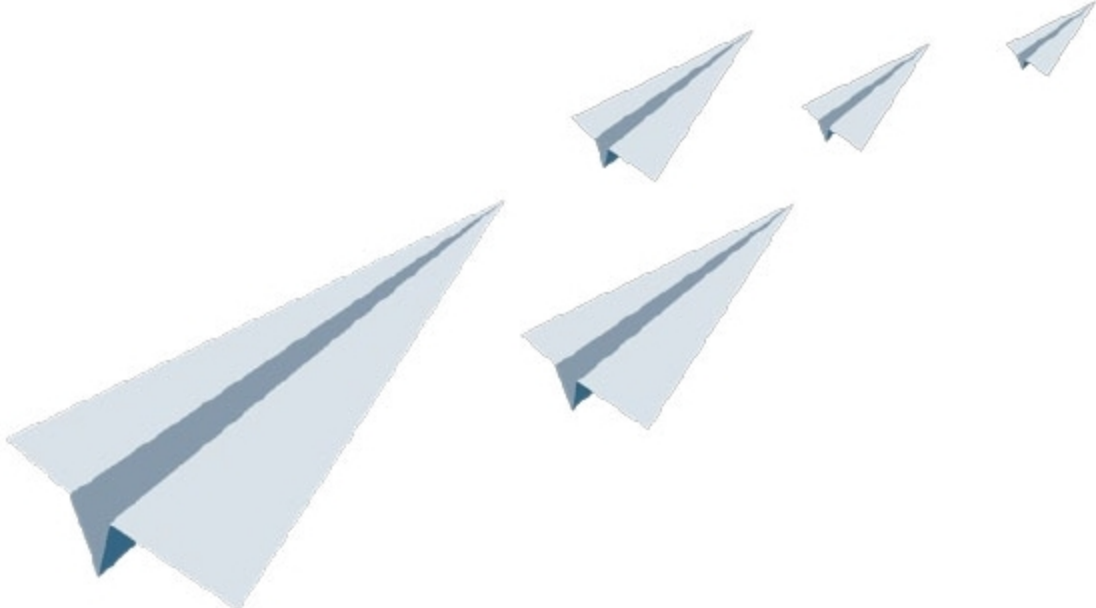




Lighttpd: A Lightweight HTTP Server *for* Embedded Systems

This article guides readers through the implementation of Lighttpd, a lightweight interactive HTTP server for embedded systems that have limited memory and storage but require real-time performance. Lighttpd is also very useful on Linux desktops.

Lighttpd is an open source Web server optimised for environments in which speed and high performance are critical. It is a viable alternative to Apache or other heavyweight Web servers. Lighttpd is standards-compliant and has built-in security as well as flexibility. The entire source code has been written in C. This comes under the BSD 'three-clause' licence. It also supports fast CGI (Common Gateway Interface) for creating dynamic content.

Prerequisites

For the system: The desktop/embedded system should run Linux. The demonstration here is based on an Ubuntu 16.04 32-bit desktop.

For the reader: The reader should know the basics of HTTP GET/POST/PUT and HTML syntax.

Installation

As the focus is on embedded platforms, the installation shown is from source code.

To start the installation, you can download the source code from <https://download.lighttpd.net/lighttpd/releases-1.4.x/lighttpd-1.4.55.tar.gz>. You can download it manually or by using the following command:

```
$ wget https://download.lighttpd.net/lighttpd/releases-1.4.x/lighttpd-1.4.55.tar.gz
```

Next, use the *tar* command to untar the tarball:

```
$ tar -zxvf lighttpd-1.4.55.tar.gz
$ cd lighttpd-1.4.55
```

After extracting the contents from the tarball, we move to the directory and do the compiling followed by the installation.

```
$ ./configure
```

 **Note:** When you are cross-compiling, you have to mention the build, host, target and other arguments to make use of your cross-compiler.

```
$ CC="/path/to/cross-compiler-gcc"
$ LD="/path/to/cross-compiler-gnu-ld"
$ ./configure --host=ppc-linux-gnu \
--build=i686-redhat-linux-gnu \
--target=powerpc-*-elf --includedir=/path/to/sysroot-for-cross
```

To get help on all the options, type the following command:

```
$ ./configure -help
```

